

Challenge 2

Adding Functionality

Exam Objectives Covered

- Identify the process required to display data in various UI elements
- Explain how to use the UnityEvent system to respond to User Input

Challenge Overview: Adding Functionality to UI Elements

Welcome to the next challenge in your SkiFree-inspired game development journey! Now that you have set up your title screen and basic UI elements, it's time to add functionality so players can load into the game scene and quit the application.

In this session, you will be working with Unity's UI Event system, which provides a prebuilt way of sending events to objects in your game based on player input. If you need a refresher on handling events in Unity, please refer to the support resource document for this session.

Step 1: Create Event Handling Script

Implement a script to handle events for the Play and Quit buttons.

Instructions:

1. Create Event Handling Script:
 - Create a new C# script named MainMenuController.
 - Attach this script to the main canvas or an empty GameObject in your title screen scene.
2. Handle Play Button Event:
 - Implement a function within MainMenuController that loads one of the game levels when the Play button is pressed.
 - Use Unity's SceneManager to load the scene.
3. Handle Quit Button Event:
 - Implement a function within MainMenuController that launches the quit check dialogue.
 - Disable the other buttons (Play, Help) on the start screen UI when the Quit button is pressed.

Step 2: Handle Quit Check Dialogue Buttons

Enhance your script to manage the quit check dialogue actions.

Instructions:

1. Initiate Quit Sequence:
 - In MainMenuController, add a function that initiates the quit game sequence when the player presses the Yes button in the quit confirmation dialog.
 - Use Unity's Application.Quit() method for this functionality.
2. Cancel Quit Sequence:
 - Add a function in MainMenuController to close the quit dialog and re-enable the main start screen UI buttons when the player presses the No button.
 - Ensure the dialog is hidden and other buttons (Play, Help) are re-enabled.

Step 3: Add Easter Egg Interaction

Create an Easter egg that changes a property of the game title when clicked.

Instructions:

1. Implement Easter Egg:
 - Add an event listener to the game title text element.
 - Choose a property to change (e.g., color, font, size) when the title is clicked.
 - Implement a function in MainMenuController that alters the selected property upon detecting a click event.

Once you've completed these steps, your game will have functional UI elements for loading the game and quitting the application, along with a fun Easter egg on the title screen.

Good luck with adding functionality to your UI! This exercise will help you develop skills in event handling, scene management, and user interaction design in Unity. Happy developing!

Note

The solution video for this challenge makes use of a free asset called DOTween to create fade animations for the UI components. If you would like to use the same solution, or examine the script included in the project, please add DOTween to your Unity project first. [You can find DOTween here.](#)

